Cardiac Output

Prodigy Anesthesia

The cardiac output (CO) is the amount of blood ejected from the heart in a single minute. If the amount of blood ejected from the heart in a single contraction (stroke volume) is known, then multiplying this value by the heart rate will provide us with the cardiac output. The cardiac output is reported in units of milliliters per minute or liters per minute.

Cardiac Output = Stroke Volume X Heart Rate

Calculate the cardiac output for a patient with a stroke volume of 122 mL and a heart rate of 54 beats per minute (BPM).

Cardiac Output = 122 mL X 54/min = 6588 mL/min

Cardiac Output Practice Problems Part A

Calculate the cardiac output for a patient whose:

1. Stroke volume = 51 mL and heart rate = 60 bpm
2. Stroke volume = 79 mL and heart rate = 71 bpm
3. Stroke volume = 31 mL and heart rate = 115 bpm
4. Stroke volume = 71 mL and heart rate = 82 bpm
5. Stroke volume = 80 mL and heart rate = 80 bpm
6. Stroke volume = 73 mL and heart rate = 151 bpm
7. Stroke volume = 90 mL and heart rate = 90 bpm
8. Stroke volume = 81 mL and heart rate = 40 bpm
9. Stroke volume = 70 mL and heart rate = 63 bpm
10. Stroke volume = 64 mL and heart rate = 75 bpm
11. Stroke volume = 45 mL and heart rate = 36 bpm
12. Stroke volume = 50 mL and heart rate = 122 bpm
13. Stroke volume = 62 mL and heart rate = 95 bpm

14. Stroke volume = 75 mL and heart rate = 48 bpm

15. Stroke volume = 109 mL and heart rate = 77 bpm

16. Stroke volume = 38 mL and heart rate = 92 bpm

17. Stroke volume = 42 mL and heart rate = 88 bpm

18. Stroke volume = 73 mL and heart rate = 101 bpm

19. Stroke volume = 111 mL and heart rate = 52 bpm

20. Stroke volume = 65 mL and heart rate = 110 bpm

21. Stroke volume = 54 mL and heart rate = 80 bpm

22. Stroke volume = 96 mL and heart rate = 55 bpm

23. Stroke volume = 115 mL and heart rate = 73 bpm

24. Stroke volume = 49 mL and heart rate = 106 bpm

25. Stroke volume = 88 mL and heart rate = 93 bpm

We learned in the section on stroke volume that the stroke volume equals the end diastolic volume (EDV) minus the end systolic volume (ESV). If this is true, then we can also calculate the cardiac output if we have these values instead.

Cardiac Output = (EDV – ESV) X Heart Rate

Calculate the cardiac output for a patient with an end diastolic volume of 148 mL, an end systolic volume of 77 mL, and a heart rate of 102 beats per minute.

Cardiac Output = (148 mL – 77 mL) X 102/min

= 71 mL X 102/min

= 7242 mL/min

Cardiac Output Practice Problems Part B

Calculate the cardiac output for patients in whom the:

1. EDV = 151 mL, the ESV = 61 mL, and the heart rate = 88/min
2. EDV = 124 mL, the ESV = 72 mL, and the heart rate = 72/min
3. EDV = 140 mL, the ESV = 65 mL, and the heart rate = 45/min
4. EDV = 132 mL, the ESV = 81 mL, and the heart rate = 61/min
5. EDV = 155 mL, the ESV = 101 mL, and the heart rate = 115/min
6. EDV = 110 mL, the ESV = 60 mL, and the heart rate = 82/min
7. EDV = 120 mL, the ESV = 52 mL, and the heart rate = 65/min
8. EDV = 131 mL, the ESV = 67 mL, and the heart rate = 76/min
9. EDV = 112 mL, the ESV = 42 mL, and the heart rate = 70/min
10. EDV = 143 mL, the ESV = 64 mL, and the heart rate = 95/min
11. EDV = 118 mL, the ESV = 70 mL, and the heart rate = 53/min
12. EDV = 133 mL, the ESV = 84 mL, and the heart rate = 97/min
13. EDV = 147 mL, the ESV = 52 mL, and the heart rate = 51/min
14. EDV = 103 mL, the ESV = 61 mL, and the heart rate = 59/min
15. EDV = 95 mL, the ESV = 35 mL, and the heart rate = 38/min
16. EDV = 126 mL, the ESV = 60 mL, and the heart rate = 52/min
17. EDV = 141 mL, the ESV = 55 mL, and the heart rate = 75/min
18. EDV = 113 mL, the ESV = 31 mL, and the heart rate = 119/min
19. EDV = 89 mL, the ESV = 42 mL, and the heart rate = 90/min
20. EDV = 115 mL, the ESV = 50 mL, and the heart rate = 63/min
21. EDV = 124 mL, the ESV = 44 mL, and the heart rate = 62/min
22. EDV = 153 mL, the ESV = 58 mL, and the heart rate = 79/min
23. EDV = 114 mL, the ESV = 48 mL, and the heart rate = 50/min
24. EDV = 125 mL, the ESV = 67 mL, and the heart rate = 44/min
25. EDV = 118 mL, the ESV = 39 mL, and the heart rate = 69/min

Cardiac Output Calculation Part A

(answers are in mL/min)

1. 3060	6. 11023	11. 1620	16. 3496	21. 4320
2. 5609	7. 8100	12. 6100	17. 3696	22. 5280
3. 3565	8. 3240	13. 5890	18. 7373	23. 8395
4. 5822	9. 4410	14. 3600	19. 5772	24. 5194
5. 6400	10. 4800	15. 8393	20. 7150	25. 8184

Cardiac Output Calculation Part B

(answers are in mL/min)

1. 7920	6. 4100	11. 2544	16. 3432	21. 4960
2. 3744	7. 4420	12. 4753	17. 6450	22. 7505
3. 3375	8. 4864	13. 4845	18. 9758	23. 3300
4. 3111	9. 4900	14. 2478	19. 4230	24. 2552
5. 6210	10. 7505	15. 2280	20. 4095	25. 5451